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Candidate geomagnetic field models for IGRF-14 and secular acceleration since 2020

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Abstract

IGRF-14 is the fourteenth generation of the International Geomagnetic Reference Field (IGRF), a spherical harmonic model of Earth's main magnetic field and its secular variation, developed through international collaboration under the auspices of the International Association of Geomagnetism and Aeronomy (IAGA). This paper describes the development, in October 2024, of candidate main field (MF) models at epochs 2020.0 and 2025.0, and a secular variation (SV) model at 2025.0, derived from Swarm satellite data, as well as the validation of the SV model using ground-based observatory measurements. Swarm data collected through May 2025 were subsequently used to update the continuous parent model from which these candidates were derived, which now spans 2014.42 to 2024.92. This model is used to conduct a retrospective assessment of SV performance and to analyze recent secular acceleration (SA) signals. Our results reveal a pronounced SA pulse centered in 2022 and provide evidence for a geomagnetic jerk in 2024, confirmed by recent observatory data from Western Europe and North America. These rapid, nonlinear core field changes have already contributed to the early degradation of the IGRF-14 SV forecast, underscoring the challenges of modeling geomagnetic field evolution and the importance of continuous satellite and ground-based observations.

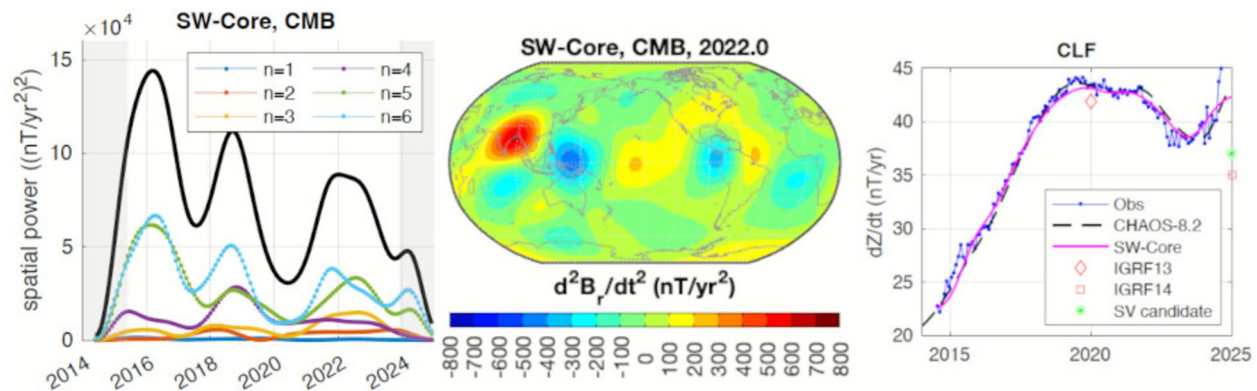
Keywords International Geomagnetic Reference Field (IGRF), Geomagnetic field modeling, Secular variation, Secular acceleration, Geomagnetic jerks, Swarm satellite mission, Geomagnetic observatories, Core dynamics

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Graphical Abstract



1 Introduction

The International Geomagnetic Reference Field (IGRF) is a spherical harmonic model of Earth's internal geomagnetic field, updated every five years and covering the period from 1900 to the present. It is developed by Working Group V-MOD of the International Association of Geomagnetism and Aeronomy (IAGA). The IGRF is widely used in scientific research, especially in geophysics, space physics, and space weather, as well as in practical applications such as mineral exploration, navigation, and satellite attitude control. The most recent version of the model is the fourteenth generation, hereafter referred to as IGRF-14, which was released in November 2024. For more information, including model equations and background material, see Beggan et al. (2025) and references therein.

IGRF-14 consists of a series of spherical harmonic models representing Earth's core magnetic field, known as the main field (MF), from epoch 1900.0 to 2025.0 at five-year intervals. It also includes a forward-looking model of the linear time variation of the geomagnetic field, referred to as the secular variation (SV), at epoch 2025.0. The MF models are expressed as spherical harmonic expansions up to degree and order 10 for the period 1900–1995, and up to degree and order 13 from 2000 onward. The geomagnetic field between model epochs is obtained by interpolating the model coefficients. The SV model at epoch 2025.0 is expanded up to degree and order 8 and provides a forecast of the field until 2030.0. Altogether, IGRF-14 spans the period from 1900.0 to 2030.0. Compared to IGRF-13, released in late 2019 and covering the period 2000.0–2025.0 (Alken et al. 2021a), IGRF-14 includes an updated MF model for epoch 2020.0, a new MF model for 2025.0, and a new SV model at epoch 2025.0. The MF and SV models at epoch

2020.0 in IGRF-13 are thus superseded by the updated models in IGRF-14.

Following a call for candidate models by IAGA, our group submitted three candidate models, one for each of the new models listed above. These models were derived from a series of parent models calculated using the most recent geomagnetic field measurements available at the time (October 2024) from the European Space Agency's (ESA) Swarm satellite mission. Swarm is a three-satellite scientific mission launched in November 2013, designed to provide high-accuracy measurements of the geomagnetic field in low-Earth orbit (Friis-Christensen et al. 2006; https://www.esa.int/Applications/Observing_the_Earth/FutureEO/Swarm). The SV candidate model was validated using independent data from ground-based magnetic observatories. While preparing this manuscript, we updated the parent models using Swarm data through May 31, 2025, which allowed us to revise the SV candidate model and perform a preliminary retrospective assessment at epoch 2025.0.

The portion of the IGRF constructed retrospectively using the best available data is referred to as the Definitive Geomagnetic Reference Field (DGRF). In IGRF-14, the DGRF series includes MF models for the epochs 1900.0 to 2020.0. For this reason, our candidate MF model at epoch 2020.0 is referred to as a DGRF candidate in what follows. The candidate MF model at epoch 2025.0 is referred to simply as the IGRF candidate.

As observed in previous generations of the IGRF and the World Magnetic Model (WMM; Chulliat et al. 2025), a similar model updated every five years, the forecasting performance of forward-looking SV models is uneven from one five-year cycle to the next and generally deteriorates over time (Chulliat et al. 2021; Alken et al. 2021b, 2021c). This decline is primarily due to a non-linear

component in the actual SV, corresponding to a non-zero second time derivative of the geomagnetic field, commonly referred to as secular acceleration (SA), which is not included in either the IGRF or WMM. The SA itself is highly variable on sub-decadal time scales, with changes originating in Earth's outer core that can occur over only a few months (Olsen & Manda 2008).

Rapid SA changes are associated with several phenomena: (1) geomagnetic jerks, which are abrupt, V-shaped changes in the SV trend at a given location, corresponding to a step change in the SA, and are well documented in ground-based observatory records (e.g., Brown et al. 2013, and references therein); (2) SA pulses, defined as sharp peaks in the global power of the SA (Chulliat et al. 2010); and (3) wave-like patterns in the radial component of the SA at the core-mantle boundary (Chulliat and Maus 2014; Finlay et al. 2015; Chulliat et al. 2015). These features have recently been attributed to various types of waves propagating within Earth's outer core, including so-called Quasi-Geostrophic Magneto-Coriolis (QG-MC) waves (Aubert and Finlay 2019; Gillet et al. 2022; Barrois and Aubert 2024).

Over the past twenty years, several SA pulses have been identified, most recently in 2018, with the strongest signals observed in the Pacific region (Finlay et al. 2020; Pavón-Carrasco et al. 2021). This pulse was followed by a geomagnetic jerk in 2019–2020 (Pavón-Carrasco et al. 2021) and significant SA changes (Bai et al. 2024). Retrospective assessments of the WMM2020 over the past five-year cycle confirmed a sharp geomagnetic jerk in 2020 at several observatories (e.g., in Honolulu, Hawaii, and Kakadu, Australia), as well as enhanced acceleration that contributed to larger forecasting errors in the South Atlantic, Central and Southeastern Asia, and Australasia regions (NCEI and BGS 2023).

These observations suggest that a new SA pulse may have occurred after 2020. To investigate this possibility, we analyzed the SA inferred from the updated parent models. The analysis revealed a distinct SA pulse centered around 2022. In addition, we identified observational evidence indicating a possible new geomagnetic jerk in 2024, which could potentially affect the accuracy of the SV forecast in IGRF-14.

The paper is organized as follows. Section 2 describes the data sources and preprocessing steps used to prepare the models presented in this study. Section 3 outlines the derivation of the models, including the candidate models submitted to IGRF-14. Section 4 details the validation of the candidate models and presents a preliminary retrospective assessment of the SV candidate. Section 5 provides an analysis of secular acceleration over the past five years, including evidence for an SA pulse centered around 2022 and a potential geomagnetic jerk in 2024.

2 Data sources and preprocessing

The models presented in this paper were derived from high-quality magnetic field measurements collected by the Swarm satellite mission. Specifically, the most recent versions of the Level 1b MAGx_LR (1 Hz) data from the Swarm Alpha and Bravo satellites were used. These data were downloaded from <https://swarm-diss.eo.esa.int>. No data from Swarm Charlie were used, as Charlie orbits within approximately 150 km of Alpha at the equator and therefore does not provide meaningful additional spatial coverage for core field modeling.

The datasets used for the different models are summarized as follows:

- IGRF-14 candidate models and their parent models: January 1, 2018, to September 30, 2024; calibration baselines 0602 (January 1 to December 11, 2018, for Alpha; January 1 to October 16, 2018, for Bravo) and 0605/0606 (all other days).
- Parent models for the preliminary retrospective assessment and the SA analysis: December 1, 2013, to May 31, 2025; calibration baselines 0602 (December 1, 2013, to December 11, 2018, for Alpha; December 1, 2013 to October 16, 2018, for Bravo), 0607 (November 1, 2024, to January 20, 2025), and 0605/0606 (all other days).

The first step of data preprocessing involved: (a) removing outliers using standard criteria, including data points with anomalously high rates of change along the orbital track (specifically, values with $|dB/dt| > 100$ nT/s or $|d^2B/dt^2| > 1$ nT/s² for at least one of the vector components), as well as data flagged for low reliability; (b) decimating the dataset by selecting one out of every ten points; and (c) selecting measurements taken during quiet geomagnetic activity and at night-side local times.

Given the abundance of high-quality Swarm data over the study period, strict selection criteria were applied to retain only data collected during periods of very low geomagnetic activity. The following criteria were applied:

- $K_p < 1+$, $K_p(t-3\text{ h}) < 1+$
- $|Dst| < 20$ nT
- $-3\text{ nT} < IMF\ B_y < 3\text{ nT}$
- $0\text{ nT} < IMF\ B_z < 5\text{ nT}$
- Local Time between 00:00 and 05:00

Here, K_p refers to the three-hourly planetary geomagnetic index, and Dst is the hourly disturbance storm-time index. $IMF\ B_y$ and $IMF\ B_z$ are components of the interplanetary magnetic field (IMF): B_y is the component perpendicular to the Sun-Earth line within the ecliptic plane,

and B_z is the north–south component perpendicular to the ecliptic plane.

The Kp index is inferred from geomagnetic field measurements at 13 subauroral observatories (e.g., Matzka et al. 2021). The Dst index is derived from measurements at four low-latitude observatories and measures the strength of the magnetospheric ring current (<https://wdc.kugi.kyoto-u.ac.jp/dstdir/>). The IMF B_y and B_z components are measured by NASA's Advanced Composition Explorer (ACE; <http://www.srl.caltech.edu/ACE>) and NOAA's Deep Space Climate Observatory (DSCOVR; <https://www.nesdis.noaa.gov/content/dscovr-deep-space-climate-observatory>). The IMF data were used to identify and exclude periods of enhanced polar ionospheric current activity.

The data were subsequently corrected for the crustal field using the MF7 model (Maus et al. 2008; <https://geomag.colorado.edu/magnetic-field-model-mf7>), and for the magnetospheric field using CHAOS-7.18 (Finlay et al. 2020) for the IGRF-14 candidate models, and CHAOS-8.2 (Kloss et al. 2025) for the models used in the retrospective and SA analyses.

3 Derivation of the models

The IGRF-14 candidate models were derived from a series of twenty-four spherical harmonic MF parent models, spanning epochs from 2018.5 to 2024.25 in 0.25-year increments. Each MF model was co-estimated with an SV model from one year of data centered on the model epoch and preprocessed as described in Sect. 2. For example, the 2018.5 MF model was derived from data collected between January 1 and December 31, 2018. The maximum degree and order were 18 for the MF models and 15 for the SV models.

Each inversion used IGRF-13 at epoch 2020.0 as the initial model and was performed using Iteratively Reweighted Least Squares (e.g., Olsen 2002) with Huber weights (Huber 1996). Outliers were identified and removed during the final iteration. Regularization was applied to both the MF and SV models: the MF models were regularized using a minimum energy norm at the core (Gubbins 1983); a single damping parameter was applied to all SV model coefficients.

Only vector data were used to derive the parent models; scalar data were excluded. This differs from more conventional approaches, which typically rely on scalar data at high latitudes to minimize the influence of field-aligned currents (e.g., Olsen 2002). We tested different configurations, including the inclusion of scalar data at high latitudes, and compared the resulting MF models with previously developed models that used scalar data at high latitudes (e.g., CHAOS-7.18) at epochs where such models were available. Differences from other models were generally similar or slightly smaller when using vector data only. For example, the root-mean-square (RMS) vector field difference between the parent model at epoch 2020 and CHAOS-7.18 is 7.2 nT, whereas it increases to 8.7 nT when scalar data above 50 degrees latitude are included. Similar differences are obtained at other epochs. This may be attributed to the application of strict geomagnetic activity selection criteria, particularly criteria aimed at minimizing perturbations in the polar regions, and to the fact that the magnetic fields generated by field-aligned currents are non-potential and thus expected to be naturally discarded by the inversion process.

Model statistics, i.e., mean and RMS misfits of the data used in the model derivation, were calculated for each parent model. Table 1 shows an example of such

Table 1 Example of model statistics for the parent model at epoch 2020.0. This model was derived from 55,086 vector triples at polar latitudes (above $\pm 50^\circ$ geomagnetic dipole latitude) and 148,484 vector triples at non-polar latitudes

	Mean misfit (nT)	RMS misfit (nT)	Weighted mean misfit (nT)	Weighted RMS misfit (nT)
B_r (all latitudes)	0.02	1.64	0.04	1.43
B_θ (all latitudes)	− 0.36	2.73	− 0.19	2.22
B_ϕ (all latitudes)	− 0.22	3.24	− 0.19	2.56
B_r (polar latitudes)	− 0.17	2.48	− 0.10	2.01
B_θ (polar latitudes)	− 0.95	3.92	− 0.54	2.88
B_ϕ (polar latitudes)	− 0.18	4.63	− 0.22	3.42
B_r (non-polar latitudes)	0.10	1.18	0.09	1.16
B_θ (non-polar latitudes)	− 0.13	2.13	− 0.07	1.95
B_ϕ (non-polar latitudes)	− 0.24	2.54	− 0.18	2.22

The number of parameters was 615. RMS stands for root mean square. Residuals are provided for the radial (r), polar (θ), and azimuthal (ϕ) components of the magnetic field

statistics for the model at epoch 2020.0. As expected, the RMS misfits are larger at polar latitudes due to stronger unmodeled external field perturbations in those regions, and the weighted RMS misfits are smaller than the unweighted ones because of the downweighting of larger residuals. At all latitudes, the RMS misfits are largest for the azimuthal and polar components, likely because of field-aligned currents (mostly at high latitudes) and residual magnetospheric fields (mostly at mid- and low latitudes). The mean misfits are generally small, suggesting a weakly biased model. The largest mean misfit is found for the polar component at high latitudes, indicating some contamination by residual magnetic fields associated with polar electrojets. Similar statistics were obtained at other model epochs.

The final step in deriving the candidate models involved fitting cubic smoothing splines to the time series of MF parent model coefficients. The smoothing parameter p was set to $p = \max(0.9 \times (1 - e^{n-12}), 0)$, where n is the spherical harmonic degree. With this choice of smoothing parameters, the spline reduces to a least-squares straight-line fit for $n \geq 12$ (i.e., we do not attempt to resolve the SA at such high degrees), while for $n \leq 10$ it behaves like a minimally smoothed natural cubic spline, allowing the secular acceleration to be well resolved. The DGRF (MF) candidate was extracted directly from the cubic splines at epoch 2020.0 for degrees 1 through 13. The IGRF MF candidate was determined by extrapolating the cubic splines for each coefficient from degree 1 to 13 to epoch 2025.0, using the spline time derivative at the epoch of the most recent parent model (2024.25). The IGRF SV candidate was calculated as the time derivative of the extrapolated cubic splines for each coefficient from degree 1 to 8 at epoch 2025.0, using a central difference approximation with a two-month interval before and after the target epoch.

The same approach was used to derive the models used in the retrospective and SA analyses, except that the parent models were derived in 1/12-year increments from epoch 2014.42 (corresponding to one year of data from December 1, 2013, to November 30, 2014) to 2024.92 (corresponding to one year of data from June 1, 2024, to May 31, 2025). The cubic smoothing splines fitted to the parent model coefficients were then used to calculate revised candidate models at epochs 2020.0 and 2025.0. They were saved as a piecewise-polynomial structure forming a continuous model over the period 2014.42–2024.92 (similar to the structure used in the CHAOS models), hereafter referred to as “SW-Core”. This continuous model was used to calculate the SV using centered six-month first differences and the SA using centered one-year second differences. The use of six-month first differences to compute the SV and one-year second

differences to compute the SA helps minimize the influence of external fields with annual and semi-annual periodicities.

4 Validation and preliminary retrospective assessment

The first step in the validation was to compare the obtained candidate models with other models developed by our group for different projects, as well as with a separate set of candidate models independently prepared using an alternative methodology. The comparisons were carried out in the spectral and spatial domains, following standard approaches for evaluating core field models (see, e.g., Alken et al. 2021b).

In addition, the IGRF SV candidate was compared with the secular variation determined at a selection of globally distributed ground-based observatories below ± 55 degrees geomagnetic latitudes (Fig. 1). We used hourly mean values from the Swarm Level 2 product “SW_OPER_AUX_OBS”, an observatory dataset quality-controlled and pre-processed by the British Geological Survey for the Swarm mission (Macmillan and Olsen 2013). For the initial model validation, we used data from 40 observatories included in the SW_OPER_AUX_OBS product released in August 2024; most observatories had data coverage up to June 30, 2024, and all had coverage until June 1, 2024. This dataset is referred to as the August 2024 dataset hereafter. For the preliminary retrospective assessment, we used data from 46 observatories included in the SW_OPER_AUX_OBS product released in May 2025, with coverage extending to early May 2025 for some observatories. This dataset is referred to as the May 2025 dataset hereafter.

The observatory data were selected for geomagnetically quiet conditions ($ap \leq 10$) and local times between 00:00 and 05:00 to reduce the influence of disturbance signals of external origin. The magnetospheric field was estimated using the CHAOS-7.18 (Finlay et al. 2020) and CHAOS-8.2 (Kloss et al. 2025) models and subtracted from the observatory data. We fitted cubic splines with knots separated by one year for the X (north), Y (east), and Z (downward vertical) components in a least-squares sense. The secular variation at each observatory was determined by subtracting the spline values at the beginning and end of a one-year window centered on the SV epoch. The spline model was then linearly extrapolated to 2025.0 using the slope at the last knot. An example of the SV determination using the May 2025 dataset is shown in Fig. 2 for the CKI (Cocos Keeling Island, Australia) magnetic observatory. Also shown in this figure are the X, Y, and Z SV values at epoch 2025.0 at CKI from the SV candidate, the SW-Core model, the IGRF-14 final SV, and the median of all SV candidates submitted to IGRF-14.

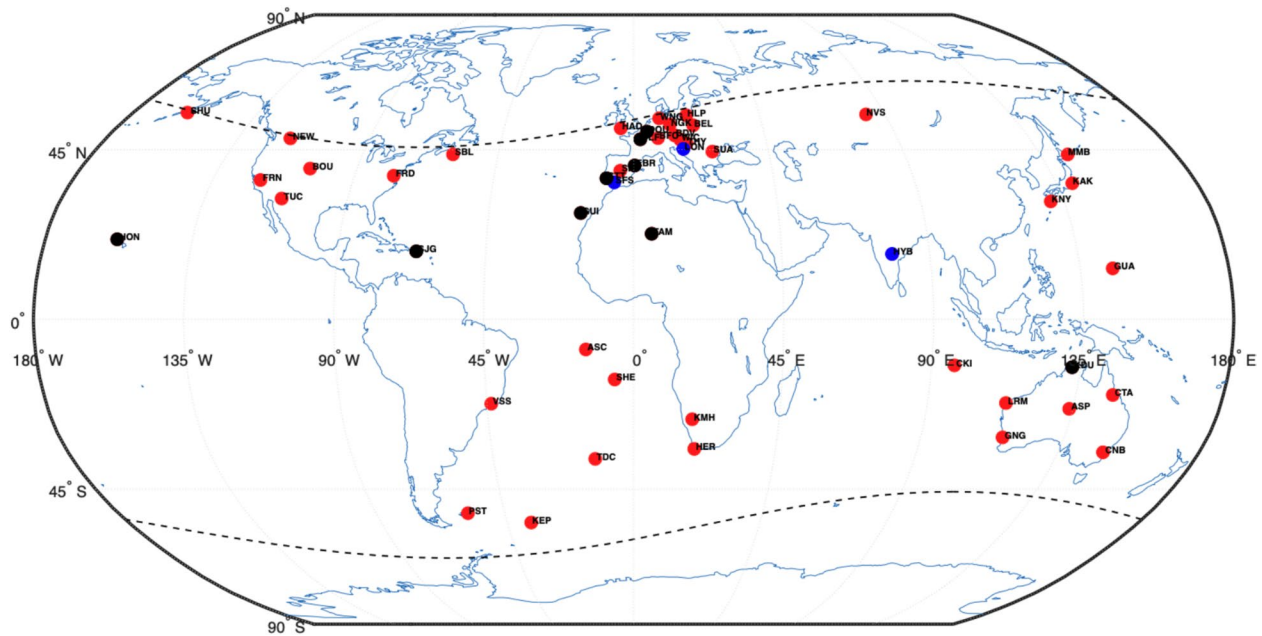


Fig. 1 Observatories used in the preliminary retrospective assessment of the IGRF SV candidate model (red dots). Of these 46 observatories, CLF, DOU, EBR, GUI, HON, KDU, SJG, STT, and TAM (black dots) were not used in the model validation, while HYB, LON, and SFS were additionally used in the validation (blue dots). The dashed lines represent ± 55 degrees geomagnetic latitudes

The differences between observatory SV and model SV at each observatory are shown in Fig. 3 for the same models as above, using the May 2025 dataset. As can be seen in this figure, the differences are generally smaller for the Y and Z components in the SW-Core SV, which incorporates more recent satellite data than the other three models. However, there is no clear overall improvement for the X component.

Detailed statistics of the differences between observatory and model SV values are presented in Table 2, both for the August 2024 (initial model validation) and May 2025 (retrospective assessment) datasets. The root-mean-square error (RMSE) is obtained by taking the square root of the sum of squares of the mean (bias) and the standard deviation of the differences. The retrospective assessment statistics confirm the improvements in Y and Z brought by SW-Core over the SV candidate, the IGRF-14 final SV and the median of all SV candidates, while showing no improvement in X. As expected, the RMSE in all three components for the SV candidate deteriorated between the initial model validation and the retrospective assessment. Compared to the IGRF-14 final SV, our SV candidate performs slightly better on Z, slightly worse on X, and significantly worse on Y. The performance of the median of all SV candidates is comparable to that of the IGRF-14 final SV. This suggests that some unmodeled, nonlinear SV occurring after the development of IGRF-14 affects the performance of all three

models at the global level, even less than one year after these models were developed.

5 Secular acceleration analysis

When analyzing rapid changes in the SA, it is helpful to downward continue the SA to the core-mantle boundary (CMB), where such changes originate. However, some precautions must be taken, as the geometrical amplification associated with downward continuation affects the higher spherical harmonic degrees exponentially more than the lower ones. This effect is even more pronounced for the SA than the MF or the SV, because the SA power spectrum at the CMB is not flat but instead increases with degree. For this reason, in the analysis presented here, the SA was either truncated at degree and order 6 or filtered using a Hann window defined by $w(n) = (1 + \cos(n\pi/N))/2$, where $N=10$ and n is the spherical harmonic degree, following Chulliat and Maus (2014). The Hann filter was applied when plotting maps of the SA or time-longitude diagrams.

The first step of the analysis involved plotting the degree-by-degree spatial power spectrum of the SA at the CMB for both the SW-Core and CHAOS-8.2 models (Fig. 4, top). Here, the spatial power spectrum represents the average squared magnitude over the sphere of radius $c=3485$ km (the Earth's core radius); see, e.g., Hulot et al. (2015). The SA from CHAOS-8.2 was computed using centered one-year second differences, as was

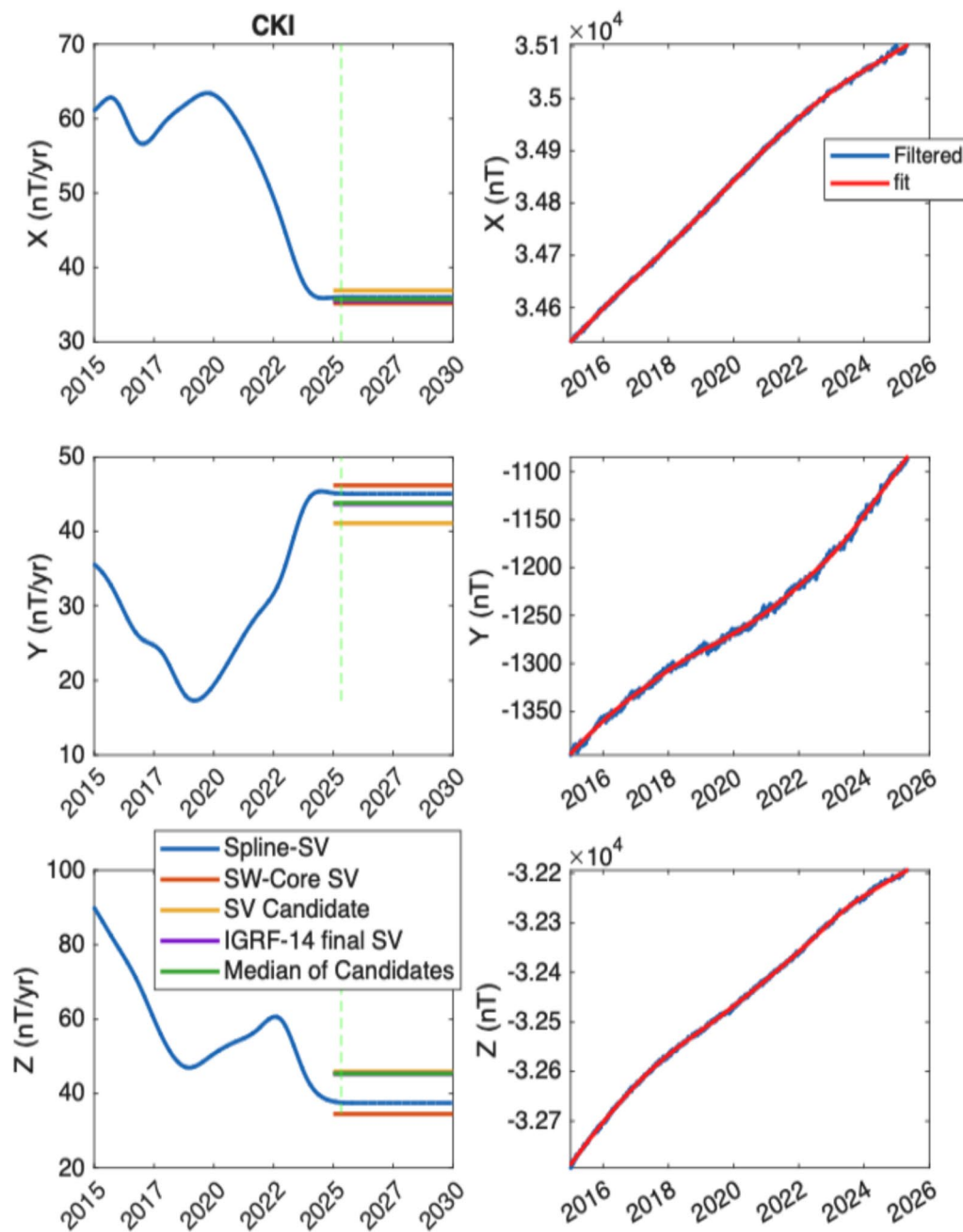


Fig. 2 Example of SV determination at the CKI (Cocos Keeling Island, Australia) magnetic observatory using the May 2025 dataset, with comparisons to the SV at CKI from various models. Left: time evolution of observatory SV (blue line) for X, Y, and Z, with SV values at 2025.0 from the SV candidate, SW-Core SV, IGRF-14 final SV, and median of all SV candidates. Because the IGRF assumes constant SV over 2025–2030, model values appear constant over 2025–2030. Right: pre-processed (filtered) CKI magnetic field data and fitted cubic splines used to infer observatory SV using centered six-month first differences

done for SW-Core. As shown in Fig. 4 (top), both models exhibit a sharp peak in the total power over degrees $n=3$ to 6, centered near epoch 2022.0. This peak is primarily associated with degrees $n=3$, 5, and 6, while no such peak is observed at degrees $n=1$, 2, or 4. Similar

temporal variations of the SA power were noted around 2006 and 2009 (Chulliat & Maus 2014), 2012 (Chulliat et al. 2015; Finlay et al. 2015), and 2016 and 2018 (Pavón-Carrasco et al. 2021), and have come to be referred to as SA pulses. The 2016 and 2018 pulses are also visible in

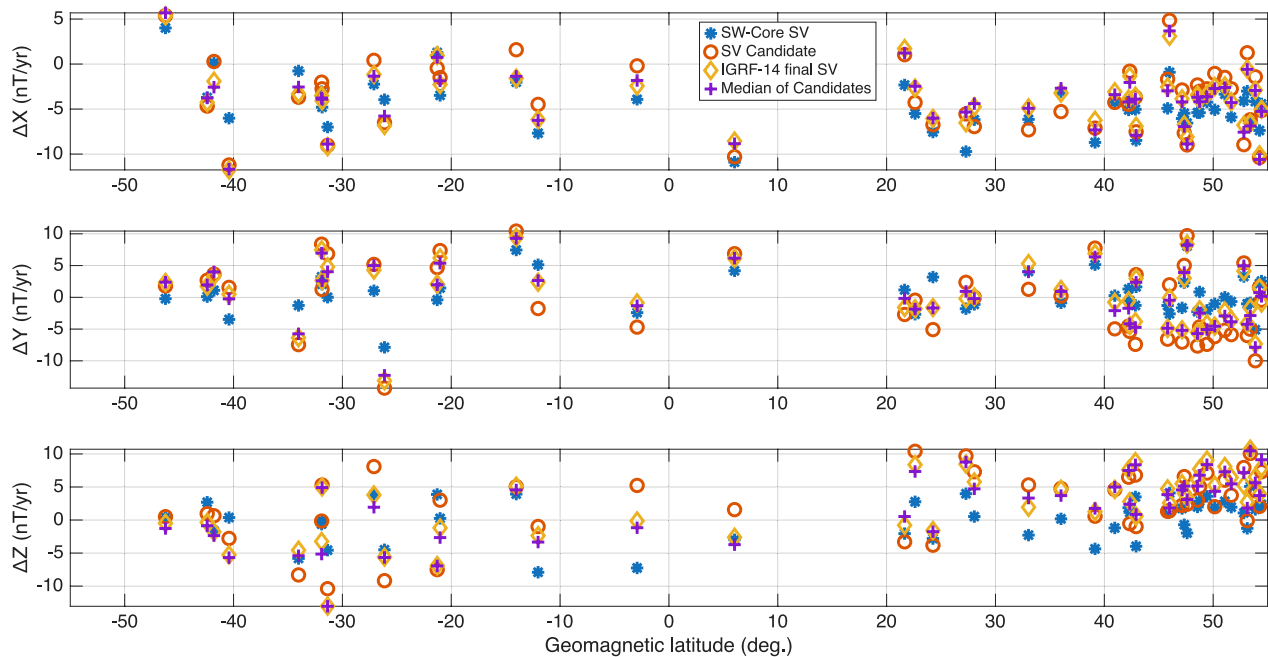


Fig. 3 Differences in X, Y, and Z between observatory SV and model SV for various models, plotted against geomagnetic latitude (± 55 degrees), using the May 2025 dataset

Fig. 4 (top). These observations indicate that another SA pulse occurred around 2022.

Due to the start date of the dataset used in SW-Core (December 1, 2013) and the methodology applied, the epoch of the first MF parent model is 2014.42, and the SA coefficients prior to 2015.42 are derived from MF models extrapolated before 2014.42 using the cubic smoothing spline algorithm (cf. Sect. 3). As a result, the SA values before 2015.42 are smoothed and considered less reliable. A similar situation applies at the end of the model's time interval: the most recent data used is from May 31, 2025, meaning the latest MF model based on a full year of data is at epoch 2024.92, and the latest reliable SA is at epoch 2023.92. For the CHAOS-8.2 model, which extends back to 1997.0, the SA can be considered reliable at the start of the interval used in this analysis. However, its latest data point is from December 31, 2024, which is five months earlier than in the case of SW-Core. Although the methodologies used to derive SW-Core and CHAOS-8.2 are different, we adopted the same criterion, requiring three years of data (1.5 years before and after a given epoch) to determine the reliability of the SA values from CHAOS-8.2. As a result, the CHAOS-8.2 SA values are considered less reliable after 2023.5. The time periods during which the SA for each model is considered less reliable are shown as gray areas in Fig. 4.

Another, much smaller SA power peak centered shortly after 2024.0 is visible in Fig. 4 (top) for the SW-Core

model. This feature is intriguing because, although it occurs during a period when the SA is considered less reliable, it follows an increase in SA power starting near the end of 2023, just before the less reliable interval. This may indicate the emergence of a new pulse beginning in late 2023, although more data will be needed to confirm this observation. Interestingly, this peak and the preceding reversal are absent from the CHAOS-8.2 plot, and both their epochs fall within the period when the CHAOS-8.2 SA is considered less reliable.

The previously referenced SA pulses over the past two decades were associated with wave-like features in the SA at low-latitudes at the CMB, characterized by intense patches of radial SA with alternating polarity. Maps of the radial component of the SA at epoch 2022.0 reveal similar structures at low latitudes, with the most intense patches located beneath South Asia (positive SA) and the Western Pacific (negative SA) for both the SW-Core and CHAOS-8.2 models (Fig. 5, top). Additional, less intense patches are visible at comparable latitudes in the Central and Western Pacific, South American, and Atlantic sectors. At epoch 2018.8 (not shown), patches of opposite polarity are observed in approximately the same locations in both SW-Core and CHAOS-8.2, consistent with the findings of Pavón-Carrasco et al. (2021), with the strongest patches located in the Western and Central Pacific regions. This longitudinal and temporal pattern of alternating-polarity radial SA patches at low latitudes,

Table 2 Statistics of the differences in X, Y, and Z between observatory SV and model SV for various models, including the mean (μ), standard deviation (σ), and root-mean-square error (RMSE)

Model	$\mu_X(\text{nT})$	$\sigma_X(\text{nT})$	$RMSE_X(\text{nT})$	$\mu_Y(\text{nT})$	$\sigma_Y(\text{nT})$	$RMSE_Y(\text{nT})$	$\mu_Z(\text{nT})$	$\sigma_Z(\text{nT})$	$RMSE_Z(\text{nT})$
(a) Initial model validation									
SV candidate	- 0.14	4.63	4.63	0.4	3.51	3.53	0.73	3.86	3.93
(b) Retrospective assessment									
SW-Core SV	- 4.63	2.72	5.37	0.42	2.98	3.01	0.17	3.15	3.15
SV candidate	- 3.75	3.78	5.33	- 0.68	5.81	5.85	2.41	4.89	5.45
IGRF-14 final SV	- 3.96	3.21	5.09	0.13	4.56	4.56	2.59	4.95	5.58
Median of all SV candidates	- 4.03	3.30	5.21	- 0.03	4.53	4.53	2.18	5.03	5.48

(a) *Model validation* refers to the analysis performed using the August 2024 dataset. (b) *Retrospective assessment* refers to the analysis performed using the May 2025 dataset

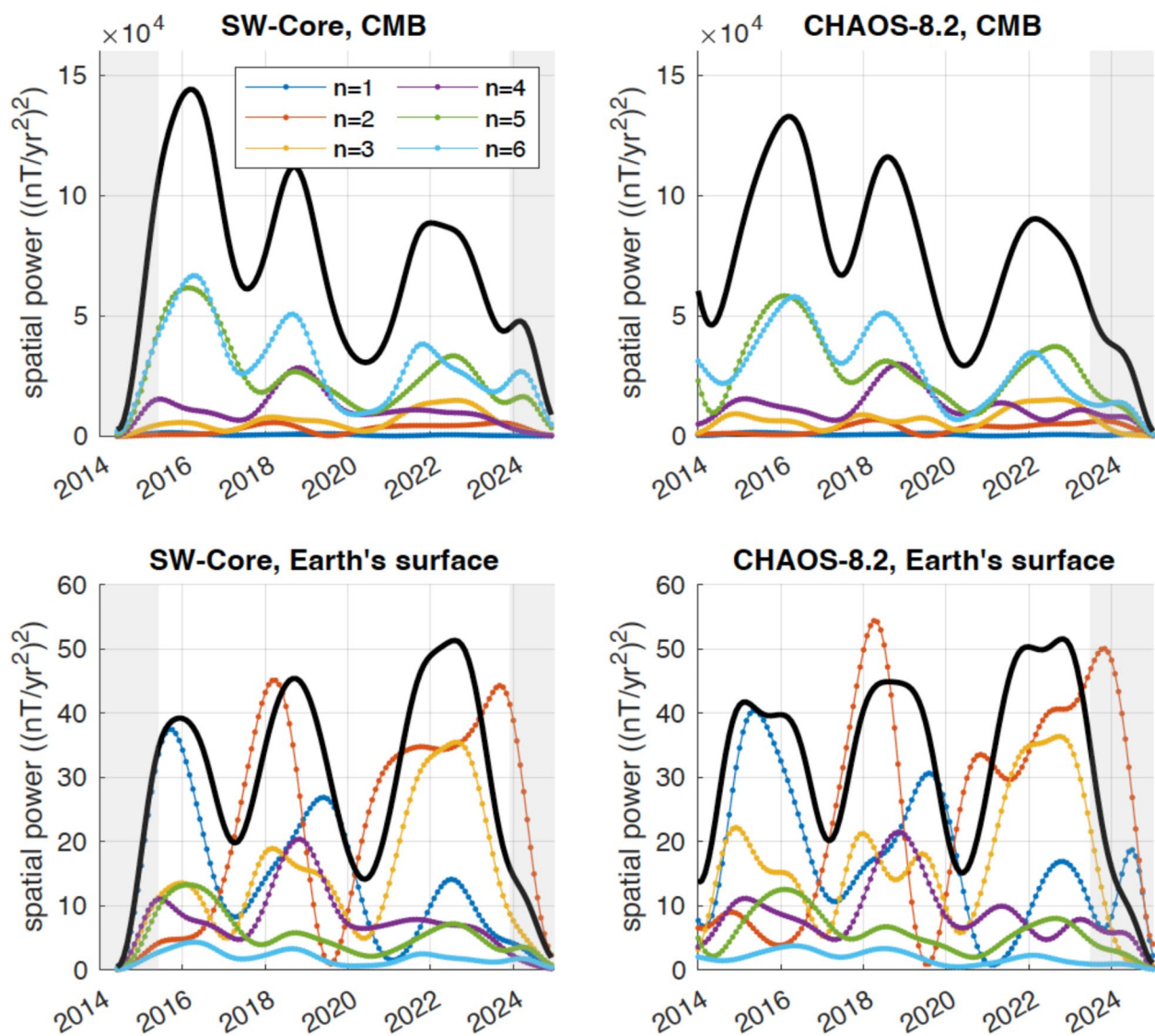


Fig. 4 Time variation of the spatial power (in $(\text{nT/yr}^2)^2$) of the first six degrees of the SA, for the SW-Core (left) and CHAOS-8.2 (right) models, at the core-mantle boundary (top) and Earth's surface (bottom). The thick black line represents the total spatial power for degrees 3 through 6. Gray areas indicate time periods during which the SA is calculated from fewer than three years of data

recurring approximately every three years, resembles earlier sequences of SA pulses, such as those observed in 2006, 2009, and 2012 in the Atlantic region (Chulliat et al. 2015).

Another way to examine these patterns is to plot time-longitude diagrams of the radial SA amplitude at the geographic equator at the CMB (Fig. 6). The Western Pacific patches of alternating polarity in 2018–19 and 2022 are prominent in these diagrams. The South Asia positive SA patch does not appear before 2023 in the time-longitude plots, as it lies slightly north of the equator. In the SW-Core model, small SA patches are visible in 2024 in the Atlantic sector, possibly indicating the onset of another

pulse. However, since they occur during the period when the SA is considered less reliable, their presence requires confirmation. These patches are preceded by a sharp reduction in SA amplitude in late 2023 at all longitudes, corresponding to the minimum in SA power observed around the same time (Fig. 4, top).

Minima in total SA power for degrees $n=3$ to 6 at the CMB often coincide with the occurrence of geomagnetic jerks at the Earth's surface, as observed in ground-based observatories (Chulliat et al., 2014), even though the lower degree SA terms ($n=1$ and 2) do not necessarily exhibit minima at the same times. This was the case during the 2020 geomagnetic jerk, for example, as shown in

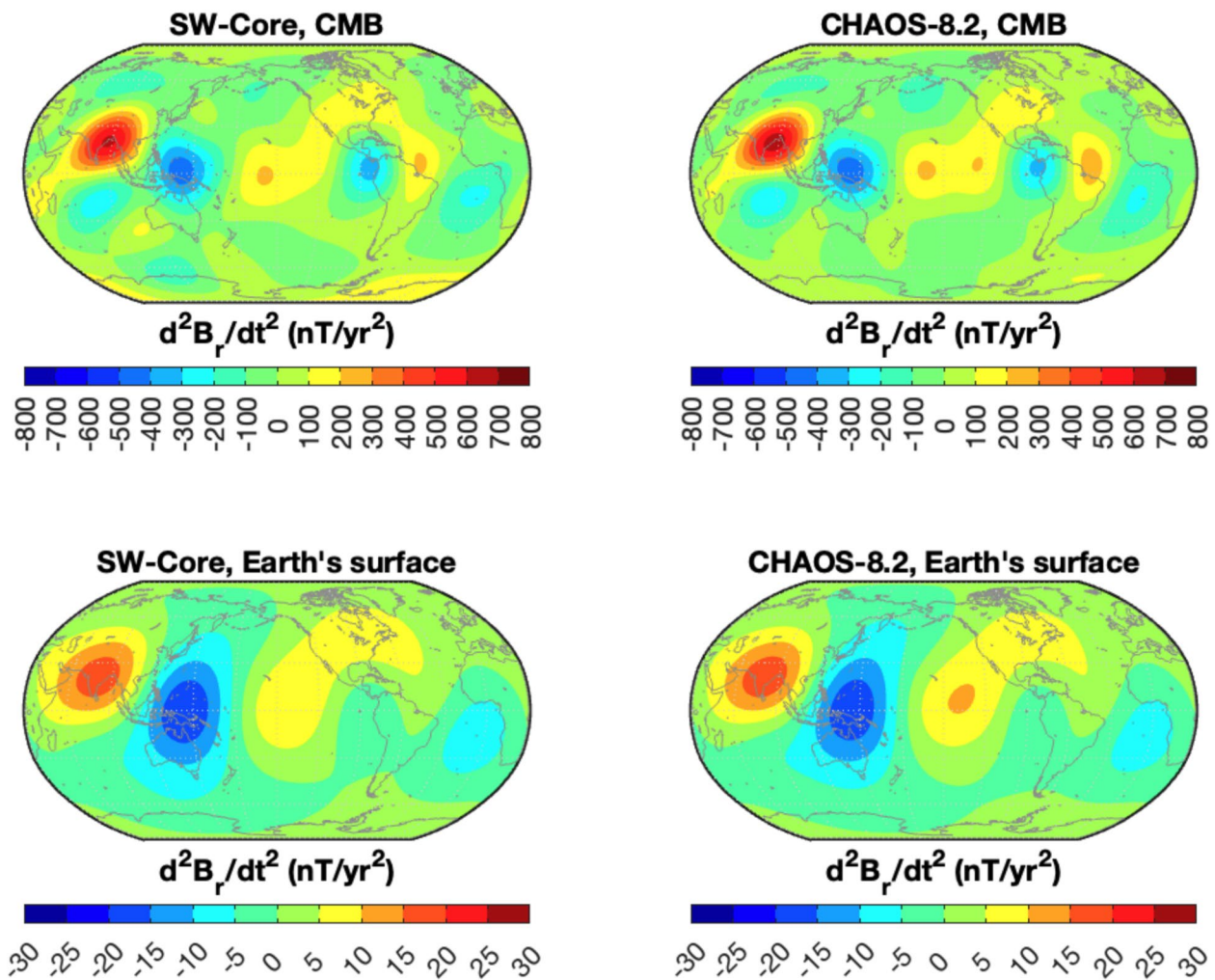


Fig. 5 Maps of the Hann-filtered SA, radial component, at epoch 2022.0 for the SW-Core (left) and CHAOS-8.2 (right) models, at the core-mantle boundary (top) and Earth's surface (bottom). The color scales are set to include the largest absolute values observed in either model over the 2014.5–2025.0 period, which occur at epoch 2018.8 for both models

Fig. 4 (bottom), when the dominant degree exhibiting a minimum around 2020 was $n=3$. During the 2022 pulse that followed, the $n=3$ degree experienced a sharp power increase, briefly reaching the same power as the $n=2$ degree. The radial SA patterns observed at the Earth's surface during this pulse confirm that degrees $n=2$ and 3 were dominant at that time (Fig. 5, bottom). These patterns also correspond to the regions of increased SV forecast error identified in the WMM2020 (NCEI & BGS, 2023).

The sharp decrease in SA power following the 2022 pulse, particularly in the SW-Core model, which is considered reliable slightly further into the future than the CHAOS-8.2 model, may signal the onset of another geomagnetic jerk around 2024. To explore this possibility, we examined the SV at observatories providing quasi-definitive data, that is, baseline-corrected measurements released on a shorter timescale than traditional definitive data (Peltier and Chulliat 2009). Such data are included in the Swarm Level 2 product

(See figure on next page.)

Fig. 6 Time-longitude diagram of the Hann-filtered SA (radial component) along the geographic equator at the core-mantle boundary for the SW-Core (top) and CHAOS-8.2 (bottom) models. Contour lines are plotted at 100 nT/yr² intervals. Longitude is in degrees. Gray areas indicate time periods during which the SA is calculated from fewer than three years of data

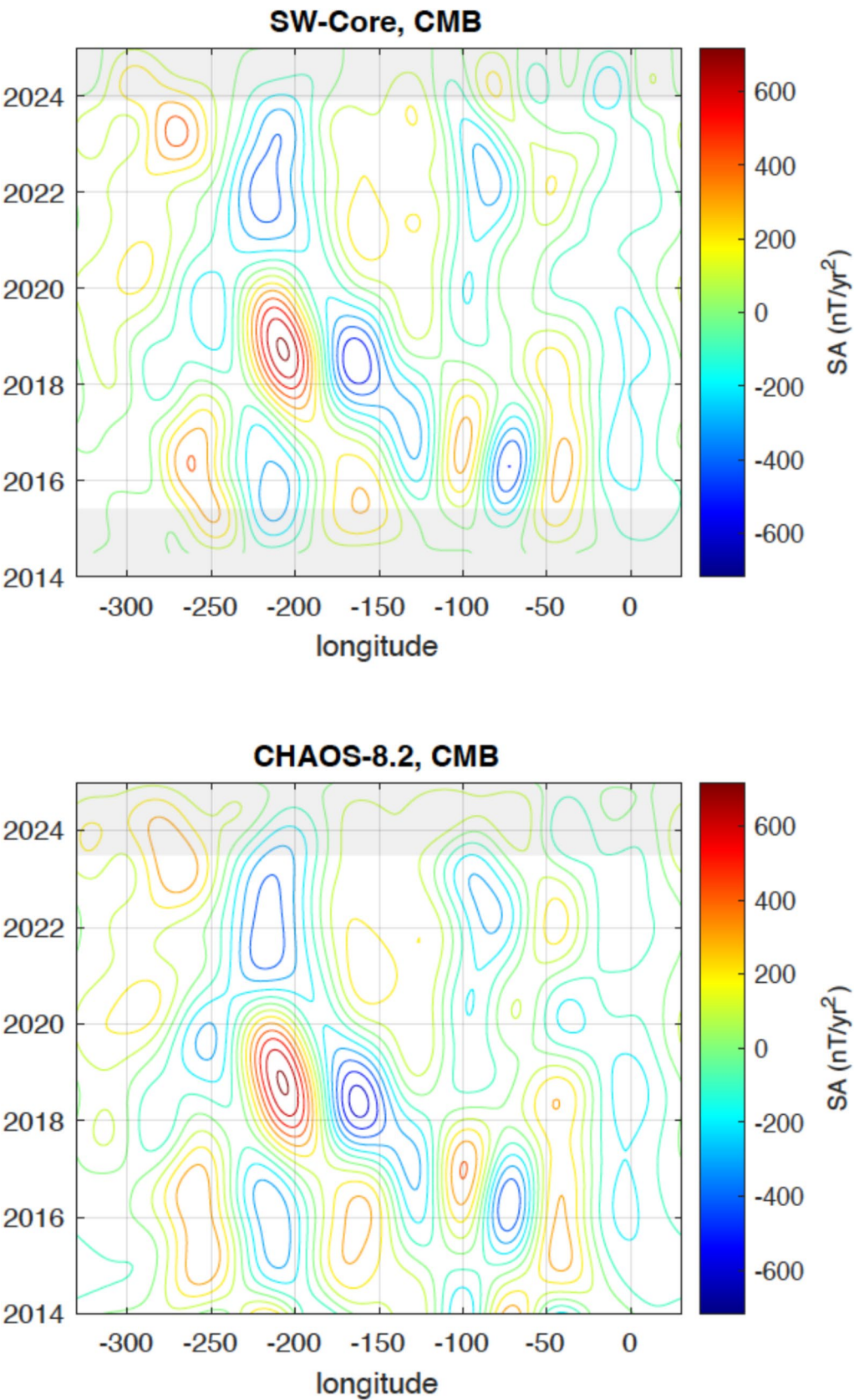


Fig. 6 (See legend on previous page.)

“SW_OPER_AUX_OBS” (Macmillan and Olsen 2013), which was already used in Sect. 4 for the model validation and preliminary retrospective assessment. Specifically, we used hourly mean values from the May 2025 version of SW_OPER_AUX_OBS.

The observatory data were pre-processed by selecting quiet-time data using the same criteria applied to the Swarm data (see Sect. 2), except that the IMF By and Bz conditions were not used. Data were also selected for periods when the solar azimuth was at least 10 degrees below the horizon. The magnetospheric field was corrected using the CHAOS-8.2 model. Monthly mean values were then computed from the selected and corrected hourly values, using daily means as an intermediate step.

The SV was derived by taking centered six-month first differences of the monthly means.

Results for selected observatories are presented in Fig. 7. These observatories are located in Western Europe (CLF: Chambon-la-Foret, France; HAD: Hartland, United Kingdom) and North America (SBL: Sable Island, Canada; BOU: Boulder, United States of America). Each site shows a V-shaped change in the SV trend around 2024.0 on the Z (downward vertical) component (only the Z component is shown, as it exhibits the strongest and clearest signal in the 2020–2025 interval). Similar changes were observed at other Western and Northern European and North American observatories (not shown). These features also appear in both the

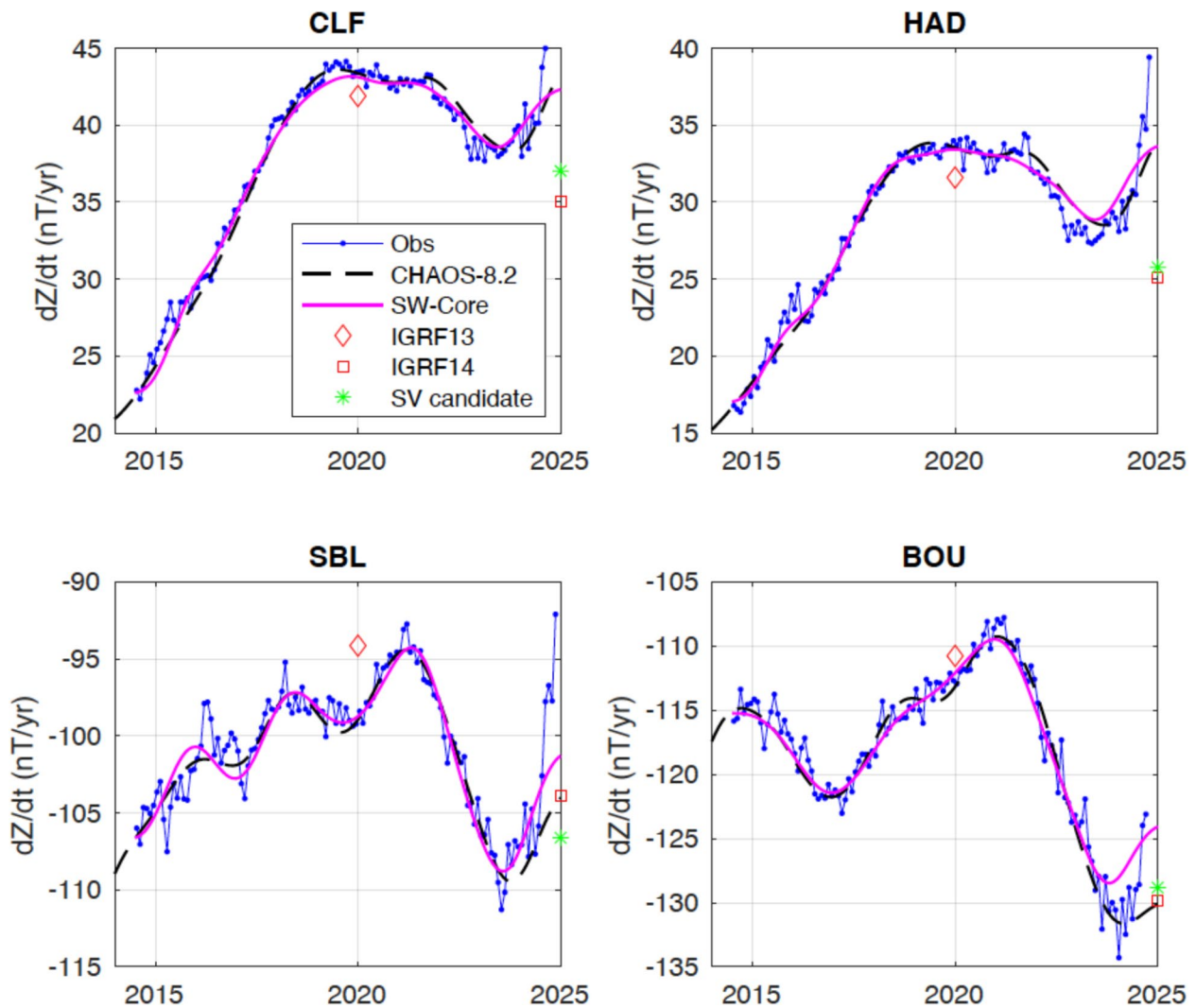


Fig. 7 Time series of the Z component of the SV at selected observatories: CLF (Chambon-la-Foret, France), HAD (Hartland, United Kingdom), SBL (Sable Island, Canada), and BOU (Boulder, United States of America). Also shown are SV predictions at each observatory from the SW-Core and CHAOS-8.2 models, as well as from IGRF13, IGRF-14, and our SV candidate model

satellite-based SW-Core model and in the satellite- and observatory-based CHAOS-8.2 model (Fig. 7), providing further evidence that these changes are real and that a geomagnetic jerk occurred in 2024 in these regions.

The nascent 2024 geomagnetic jerk was not captured by our candidate SV model or the IGRF-14 final SV in Western Europe (see the CLF and HAD plots in Fig. 7) and was only partially captured by these models in North America (SBL and BOU plots). The associated modeling errors are expected to contribute, at least in part, to the degradation in the performance of both our candidate SV and the IGRF-14 SV observed in the preliminary retrospective assessment presented in Sect. 4. A similar situation occurred shortly after the development of IGRF-13, due to the 2020 geomagnetic jerk and subsequent 2022 SA pulse, as seen in the SBL and BOU plots in Fig. 7. At that time, the Z-component errors over the five-year IGRF cycle became larger in North America compared to Western Europe.

6 Conclusion

In this paper, we provided detailed technical information on how the DGRF, IGRF MF and IGRF SV candidate models submitted by our group to IGRF-14 were developed and validated. All three models were inferred from magnetic field vector measurements collected by the Swarm Alpha and Bravo satellites from January 1, 2018, to September 30, 2024. We later extended the dataset backward to December 1, 2013, and forward to May 31, 2025. This allowed us to compute an updated set of parent models, from which a continuous model over the period 2014.42–2024.92, referred to as “SW-Core”, was inferred.

The SV at 2025.0 inferred from SW-Core, as well as observatory data collected until May 2025, were used to perform a preliminary retrospective assessment of our IGRF-14 SV candidate and the IGRF-14 final SV. We found that the performance of both models has already deteriorated in all three components, especially the Y and Z components, even less than one year after the model were developed, suggesting the recent occurrence of non-linear SV with significant intensity and geographical extent.

Analysis of the secular acceleration from SW-Core since 2020 revealed the signature of a new SA pulse at the CMB, centered in 2022, immediately following the 2020 geomagnetic jerk and the 2016 and 2018 pulses previously reported by others. The analysis also revealed low-latitude wave-like patterns in the radial SA, similar to those observed in earlier pulses attributed to waves in the core. It further suggested that the 2022 SA pulse might be followed by another

geomagnetic jerk in 2024. The occurrence of a jerk in 2024 was indeed confirmed by recent observatory data in Western European and North America. However, the true geographical coverage and intensity of this event remain to be determined and will require more data. Notably, it is currently unclear whether a geomagnetic jerk has recently occurred at low latitudes.

The findings of this paper have implications for the forecasting accuracy of IGRF-14. It appears likely that the performance of the IGRF-14 SV will continue to deteriorate in the near future, particularly in regions affected by the 2024 geomagnetic jerk, including Western Europe and North America. If the timescales of the non-linear SV remain similar to those observed over the past two decades, it could take two or three years before the full extent of the geomagnetic jerk and its impact on IGRF-14 performance are fully understood.

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Author contributions

AC led the development of the candidate models, conducted the analysis of secular acceleration, and wrote the manuscript. MN contributed to data analysis and performed the observatory-based validation of the models. SC contributed to data analysis and independently developed an alternative set of candidate models, which was used for validation purposes. All authors reviewed, edited, and approved the final manuscript.

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Availability of data and materials

The IGRF-14 model coefficients and associated software are publicly available at <https://www.ncei.noaa.gov/products/international-geomagnetic-reference-field>. The candidate models submitted by our group are available at the IAGA Working Group V-MOD GitHub repository: <https://github.com/IAGA-VMOD/IGRF14eval>. The continuous parent model used in this study is available upon request. Swarm satellite data and ground-based observatory magnetic field measurements are accessible via the ESA Swarm Data Access portal: <https://swarm-diss.esa.int>. Geomagnetic activity indices used include the Kp and ap indices from GFZ (<https://kp.gfz.de/en/data>), the Dst index from Kyoto WDC (<https://wdc.kugi.kyoto-u.ac.jp/dstdir/>), and solar wind parameters from NASA/SPDF OMNIWeb (https://spdf.gsfc.nasa.gov/pub/data/omni/omni_cdaweb/hourly/).

Declarations

Competing interests

The authors declare that they have no competing financial or non-financial interests.

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References

- Alken P, Thébault E, Beggan CD, Aubert J, Baerenzung J, Brown WJ, Canet E, Chulliat A, Finlay CC, Gillet N, Hamilton B, Huder L, Hulot G, Kloss C, Lesur V, Livermore PW, Metman MC, Olsen N, Pavón-Carrasco FJ et al (2021a) International geomagnetic reference field: the thirteenth generation. *Earth Planets Space* 73:49. <https://doi.org/10.1186/s40623-020-01288-x>
- Alken P, Thébault E, Beggan CD, Aubert J, Baerenzung J, Brown WJ, Canet E, Chulliat A, Finlay CC, Gillet N, Hamilton B, Huder L, Hulot G, Kloss C, Lesur V, Livermore PW, Metman MC, Olsen N, Pavón-Carrasco FJ et al (2021b) Evaluation of candidate models for the 13th generation International Geomagnetic Reference Field. *Earth Planets Space* 73:48. <https://doi.org/10.1186/s40623-020-01281-4>
- Alken P, Chulliat A, Nair M (2021c) NOAA/NCEI and University of Colorado candidate models for IGRF-13. *Earth Planets Space*. <https://doi.org/10.1186/s40623-020-01313-z>
- Aubert J, Finlay CC (2019) Geomagnetic jerks and rapid hydromagnetic waves focusing at Earth's core surface. *Nat Geosci* 12(5):393–398. <https://doi.org/10.1038/s41561-019-0355-1>
- Bai C, Zhang X, Liu L, Liu J (2024) Dynamic evolution of amplitude and position of geomagnetic secular acceleration pulses since 2000. *Front Earth Sci* 12:1383149. <https://doi.org/10.3389/feart.2024.1383149>
- Barrois O, Aubert J (2024) Characterization of hydromagnetic waves propagating over a steady, non-axisymmetric background magnetic field. *Proc R Soc Math Phys Eng Sci* 480(2280):20240067. <https://doi.org/10.1098/rspa.2024.0067>
- Beggan CD, Kloss C, et al (2025) International Geomagnetic Reference Field: the fourteenth generation. In: preparation for Earth, Planets and Space.
- Brown WJ, Mound JE, Livermore PW (2013) Jerks abound: an analysis of geomagnetic observatory data from 1957 to 2008. *Phys Earth Planet Inter* 223:62–76. <https://doi.org/10.1016/j.pepi.2013.06.001>
- National Centers for Environmental Information (U.S.), and British Geological Survey (2023) State of the geomagnetic field 2023. <https://doi.org/10.25923/cww1-sc35>
- Chulliat A, Maus S (2014) Geomagnetic secular acceleration, jerks, and a localized standing wave at the core surface from 2000 to 2010. *J Geophys Res Solid Earth* 119(2):1531–1543. <https://doi.org/10.1002/2013JB010604>
- Chulliat A, Thébault E, Hulot G (2010) Core field acceleration pulse as a common cause of the 2003 and 2007 geomagnetic jerks. *Geophys Res Lett* 37:L07301. <https://doi.org/10.1029/2009GL042019>
- Chulliat A, Alken P, Maus S (2015) Fast equatorial waves propagating at the top of the Earth's core. *Geophys Res Lett* 42(9):3321–3329. <https://doi.org/10.1002/2015GL064067>
- Chulliat A, Brown W, Alken P, Macmillan S, Panizza M (2021) Modeling Earth's ever-shifting magnetism. *Eos*. <https://doi.org/10.1029/2021EO153457>
- Chulliat A, Brown W, Nair M, Gomez Perez N, Young LY, Watson C., Boneh N, Beggan C, Meyer B, Panizza M (2025) The US/UK world magnetic model for 2025–2030: technical report. National Centers for Environmental Information, NOAA. <https://doi.org/10.25923/prbc-s316>
- Finlay CC, Olsen N, Tøffner-Clausen L (2015) DTU candidate field models for IGRF-12 and the CHAOS-5 geomagnetic field model. *Earth Planets Space* 67(1):114. <https://doi.org/10.1186/s40623-015-0274-3>
- Finlay CC, Kloss C, Olsen N, Hammer M, Tøffner-Clausen L, Grayver A, Kuvshinov A (2020) The CHAOS-7 geomagnetic field model and observed changes in the South Atlantic Anomaly. *Earth Planets Space* 72:156. <https://doi.org/10.1186/s40623-020-01252-9>
- Friis-Christensen E, Lühr H, Hulot G (2006) *Swarm*: a constellation to study the Earth's magnetic field. *Earth Planets Space* 58(4):351–358. <https://doi.org/10.1186/BF03351933>
- Gillet N, Gerick F, Angappan R, Jault D (2022) Satellite magnetic data reveal interannual waves in Earth's core. *Proc Natl Acad Sci* 119:e2115258119. <https://doi.org/10.1073/pnas.2115258119>
- Gubbins D (1983) Geomagnetic field analysis—I. stochastic inversion. *Geophys J Int* 73(3):641–652. <https://doi.org/10.1111/j.1365-246X.1983.tb03336.x>
- Huber PJ (1996) Robust statistical procedures. Society for Industrial and Applied Mathematics
- Hulot G, Sabaka TJ, Olsen N, Fournier A (2015) The present and future geomagnetic field. In: Schubert G (ed) *Treatise on geophysics*, vol 5, 2nd edn. Elsevier, pp 33–78. <https://doi.org/10.1016/B978-0-444-53802-4.00096-8>
- Kloss C, Finlay CC, Olsen N, Tøffner-Clausen L, Gillet N, Grayver A (2025) CHAOS-8 geomagnetic field model (8.2). Zenodo. 10.5281/zenodo.14893049
- Macmillan S, Olsen N (2013) Observatory data and the Swarm mission. *Earth Planets Space* 65(11):1355–1362. <https://doi.org/10.5047/eps.2013.07.011>
- Matzka J, Stolle C, Yamazaki Y, Bronkalla O, Morschhauser A (2021) The geomagnetic Kp index and derived indices of geomagnetic activity. *Space Weather* 19:e2020SW2641. <https://doi.org/10.1029/2020SW002641>
- Maus S, Yin F, Lühr H, Manoj C, Rother M, Rauberg J, Michaelis I, Stolle C, Müller RD (2008) Resolution of direction of oceanic magnetic lineations by the sixth-generation lithospheric magnetic field model from CHAMP satellite magnetic measurements. *Geochem Geophys Geosyst* 9:Q07021. <https://doi.org/10.1029/2008GC001949>
- Olsen N (2002) A model of the geomagnetic field and its secular variation for epoch 2000 estimated from Ørsted data. *Geophys J Int* 149(2):454–462. <https://doi.org/10.1046/j.1365-246X.2002.01657.x>
- Olsen N, Manda M (2008) Rapidly changing flows in the Earth's core. *Nat Geosci* 1(6):390–394. <https://doi.org/10.1038/ngeo203>
- Pavón-Carrasco FJ, Marsal S, Campuzano SA, Torta JM (2021) Signs of a new geomagnetic jerk between 2019 and 2020 from Swarm and observatory data. *Earth Planets Space* 73:175. <https://doi.org/10.1186/s40623-021-01504-2>
- Peltier A, Chulliat A (2010) On the feasibility of promptly producing quasi-definitive magnetic observatory data. *Earth Planets Space* 62(2):e5–e8. <https://doi.org/10.5047/eps.2010.02.002>

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